A Bayesian Critique of Rank-Based Methods for Surrogate Marker Evaluation

Pietro Carlotti[1] and Layla Parast[2]

[1]Department of Statistics and Data Sciences, University of Texas at Austin, 105 E 24th St D9800, Austin, TX 78705, pietro.carlotti@utexas.edu

[2]Department of Statistics and Data Sciences, University of Texas at Austin, 105 E 24th St D9800, Austin, TX 78705

Abstract

Surrogate markers are often employed in clinical trials to replace primary outcomes that may be difficult, expensive, or time-consuming to measure directly. These markers can accelerate the evaluation of new treatments, provided they reliably capture the causal relationship between treatment and true clinical benefit. Parast et al. (2024) recently proposed a rank-based approach for evaluating surrogate markers, characterized by its nonparametric nature and minimal assumptions. While this method is useful in small-sample model-agnostic settings, it has several limitations, including a lack of clear causal interpretation, low statistical power, and insufficient robustness to different data-generating mechanisms. In this paper, we propose a Bayesian approach that addresses these shortcomings by focusing on causal treatment effect estimands and, in doing so, improves power through covariate adjustment. We demonstrate the advantages of our proposed method through a simulation study designed to highlight gains in both accuracy and power.

Keywords: Bayesian Inference; Surrogate Markers; Clinical Trials; Causal Inference.

1 Introduction

Surrogate markers are often employed in clinical trials to replace primary outcomes that may be difficult, costly, or time-consuming to measure. These markers can accelerate the evaluation of new treatments, provided they reliably capture the causal relationship between treatment and true clinical benefit. For example, Lavine et al. (2010) conducted a clinical trial evaluating various treatments for children with nonalcoholic fatty liver disease, in which the primary outcome, a measure of liver function, was obtained via liver biopsy, a very invasive procedure. Instead, this study examined treatment effectiveness using a surrogate marker, change in alanine aminotransferase (ALT), a blood-based enzyme that reflects liver function, thus substantially reducing patient burden. Notably, in the United States, new treatments can be approved early based on demonstrated effectiveness on a surrogate marker via the Food and Drug Administration's (FDA) Accelerated Approval Program, established in 1992 (FDA, 1992). Many drugs have been and continue to be approved through this mechanism (FDA, 2025).

A valid surrogate can substantially reduce trial burden, duration, and cost while still providing meaningful information about treatment efficacy, thereby enabling more timely access to effective treatments. However, establishing *surrogate validity* is a challenging statistical problem, requiring methods that are both robust and interpretable. On this account, surrogate evaluation has been an active area of research, particularly because the FDA does not provide a formal statistical definition of a valid surrogate. A recent review by Elliott (2023) summarizes the state of proposed and currently used methods, including the proportion of treatment effect explained (PTE) framework (Freedman et al., 1992; Wang and Taylor, 2002; Parast et al., 2016; Parast, 2025), the principal stratification framework (Frangakis and Rubin, 2002; Gilbert and Hudgens, 2008), and the meta-analytic framework (Buyse and Molenberghs, 1998; Burzykowski et al., 2005; Huang and Gilbert, 2011).

Unfortunately, these methods often perform poorly in small trials, especially when using robust techniques such as kernel-based estimation within the PTE framework. To address this, Parast et al. (2024) proposed a rank-based approach for evaluating surrogate markers in such settings. Although attractive for its nonparametric nature and minimal assumptions, this method has notable limitations, including unclear causal interpretation for its target estimands, low power, and limited robustness across data-generating mechanisms.

In this paper, we improve upon this existing approach by proposing an imputation-based Bayesian alternative that extends the rank-based framework of Parast et al. (2024). Our approach is grounded in Bayesian causal inference, which treats unobserved potential outcomes as latent variables and samples them from their posterior predictive distributions (Li et al., 2023). By using this perspective, our method not only provides a causal interpretation but also allows for covariate adjustment, improved efficiency, and a flexible framework for inference. The paper is organized as follows. In Section 2, we review the existing rank-based approach for surrogate evaluation, outline its limitations, and emphasize how the choice of treatment effect estimands can influence conclusions about surrogate validity. In Section 3, we present our Bayesian methodology, including two parametric models for the data-generating process and a new criterion for the surrogate validation threshold. In Section 4, we present a simulation study comparing the performance of our proposed method to the existing approach. Finally, in Section 6 we discuss potential future extensions.

2 Methods

2.1 Notation

Let Y denote the primary outcome and S the surrogate marker. Let $Z \in \{0, 1\}$ be the treatment indicator, assigned at random, where $Z = 1$ indicates treatment and $Z = 0$

control. Without loss of generality, assume higher values of Y and S correspond to greater clinical benefit. We adopt the potential outcomes framework to define causal effects (Rubin, 1974). For each unit $i \in \{1, \ldots, n\}$, let Y_{gi} and S_{gi} denote the potential outcomes for the primary and surrogate under treatment $g \in \{0, 1\}$. The potential outcomes for unit i are

$$P_i = (Y_{0i}, Y_{1i}, S_{0i}, S_{1i})^\top.$$

We assume that for each unit i, the vector of potential outcomes P_i is independently drawn from a joint distribution F_i, which may differ across units, and that the treatment assignment Z_i is independently drawn from a Bernoulli distribution, with $Z_i \perp\!\!\!\perp P_i$ by design. Only the potential outcomes corresponding to the observed treatment assignment are observed. That is, each unit i contributes either (i) $P_{1i} = (Y_{1i}, S_{1i})^\top$ if $Z_i = 1$, or (ii) $P_{0i} = (Y_{0i}, S_{0i})^\top$ if $Z_i = 0$. Then, the hierarchical process that generates the data is summarized as follows:

$$\begin{aligned} (Y_i, S_i)^\top &= \begin{cases} P_{1i}, & \text{if } Z_i = 1, \\ P_{0i}, & \text{if } Z_i = 0, \end{cases} \quad i = 1, \ldots, n, \\ P_i &\overset{\text{ind.}}{\sim} F_i, \quad i = 1, \ldots, n, \\ Z_i &\overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p), \quad i = 1, \ldots, n, \end{aligned} \tag{1}$$

where p is the known treatment assignment probability (e.g., $p = 0.5$ for a balanced trial).

In the next section, we describe the existing rank-based approach for evaluating a surrogate marker and call attention to its key limitations.

2.2 Existing Frequentist Rank-Based Approach

The main idea of Parast et al. (2024) is that if S is a valid surrogate for Y, then the treatment Z should exert a comparable effect, on a probabilistic scale, on both. Their method focuses on small-sample settings, where nonparametric kernel methods are infeasible and parametric models are difficult to validate. Motivated by this, they introduced the following probabilistic index:

$$U_Y = \mathbb{P}(Y_{1i} > Y_{0j}) + \frac{1}{2}\,\mathbb{P}(Y_{1i} = Y_{0j}), \tag{2}$$

as a measure of the treatment effect on the primary outcome Y, where i and j index two different units. This quantity is often referred to as the Mann-Whitney parameter (Mann and Whitney, 1947; Thas et al., 2012; Hoeffding, 1992; Buyse, 2010), as it is the estimand targeted by the corresponding hypothesis test. In this work, we restrict our attention to continuous outcomes Y, thereby excluding the possibility of ties in Equation (2). Therefore, a positive treatment effect on Y is concluded if $U_Y > 0.5$. The same reasoning applies to the surrogate S, with U_S defined analogously. Then, the discrepancy in treatment effects is measured by their difference $\delta = U_Y - U_S$. Based on this definition, Parast et al. (2024) propose that, if S is a valid surrogate for Y, δ should be small, leading to the following hypothesis test:

$$\begin{cases} H_0 : \delta \geq \varepsilon, \\ H_1 : \delta < \varepsilon, \end{cases} \tag{3}$$

where ε is a pre-specified threshold for surrogate validation.

To estimate the parameter of interest δ, Parast et al. (2024) compute $\widehat{U}_Y$ and $\widehat{U}_S$ using

plug-in estimators, commonly known as the Mann–Whitney U-statistics:

$$\begin{aligned}\widehat{U}_Y &= \frac{1}{n_1 n_0}\sum_{i=1}^{n_1}\sum_{j=1}^{n_0}\mathbb{1}(Y_i > Y_j),\\ \widehat{U}_S &= \frac{1}{n_1 n_0}\sum_{i=1}^{n_1}\sum_{j=1}^{n_0}\mathbb{1}(S_i > S_j),\end{aligned} \tag{4}$$

where n_1 and n_0 are the treated and control sample sizes, and then estimate δ by $\widehat{\delta} = \widehat{U}_Y - \widehat{U}_S$. A confidence interval follows from the central limit theorem for U-statistics and closed-form variance estimators for correlated U-statistics. Parast et al. (2024) conclude that S is valid if the upper bound of this interval is less than the pre-specified threshold ε.

2.3 Limitations of the Existing Approach

The nonparametric approach of Parast et al. (2024) provides a method for surrogate evaluation in small-sample settings, but it suffers from several notable limitations. First, it is worth questioning whether the treatment effect quantities considered truly capture the causal targets. In fact, in the causal inference literature rank-based statistics have been criticized for lacking clear causal interpretation (Imbens and Rubin, 2015). For example, the Mann–Whitney test assesses whether two variables are equal in distribution, but this condition is not necessarily informative about causal effects. To illustrate this point, consider that since U_Y represents the probability that a randomly selected treated unit has a higher Y than an independently selected control, it measures global distributional differences between groups. But is this the parameter we truly want to characterize? A more compelling target may be

$$V_Y = \mathbb{P}(Y_{1i} > Y_{0i}), \tag{5}$$

which represents the probability that a randomly selected unit has a better outcome Y under treatment than under control. If we define V_S analogously for the surrogate S, then we can measure the discrepancy in treatment effects via $\theta = V_Y - V_S$. Notably, U_Y and U_S are not equivalent to V_Y and V_S, as discussed in detail by Fay et al. (2018). This difference propagates directly to the discrepancy parameters δ and θ. As we illustrate in Section 2.4, they can differ substantially even under relatively simple and well-behaved data-generating mechanisms, and therefore the choice between them may lead to opposing conclusions.

Secondly, the assumptions underlying the Mann–Whitney statistic may be too restrictive. In particular, the test assumes potential outcomes for each observation are independent and identically distributed. The identical distribution assumption is often unrealistic, as potential outcomes are typically influenced by covariates that vary across units.

Moreover, the numerical studies in Parast et al. (2024) show that, while their estimation approach achieves minimal bias across various settings, the resulting test evaluating the surrogate suffers from low power. Approaches that prioritize higher power, even if they require some parametric assumptions, may better detect valid surrogates, especially in small-sample settings. Along these lines, leveraging baseline covariates is a natural strategy to increase power, as covariate adjustment is known to improve both power and efficiency in other settings by reducing residual outcome variability unrelated to treatment (Zhang et al., 2008; Colantuoni and Rosenblum, 2015; Zhang and Ma, 2019). However, the rank-based framework of Parast et al. (2024) does not easily allow for covariate adjustment. Incorporating baseline information would likely require a rank regression–type approach, representing a substantial methodological departure from the original framework.

Hence, in Section 3, we describe our Bayesian imputation-based methodology addressing these limitations. In particular, our approach focuses on the causal treatment effect quantities V_Y and V_S, improves power, and allows incorporation of baseline covariates. Before

doing so, in the following section, we detail the difference between the discrepancy parameters; this section may be skipped without loss of continuity by readers less interested in these subtleties.

2.4 Comparison between Discrepancy Parameters

As noted in Section 2.3, the parameters U_Y and V_Y are not equivalent, and this difference propagates directly to the discrepancy parameters δ and θ. In practice, this means that, under the same data-generating mechanism, the two discrepancy parameters can take substantially different values, potentially leading to opposing conclusions about surrogate validity. To illustrate this point, it is useful to consider the analytical forms of δ and θ in a simple setting where the potential outcomes are drawn from a Gaussian distribution with parameters that depend on a baseline covariate X:

$$\begin{aligned} P_i &\overset{\text{ind.}}{\sim} \mathcal{N}_4\big(\mu^{X_i},\ \Sigma^{X_i}\big), \quad i = 1, \ldots, n, \\ X_i &\overset{\text{i.i.d.}}{\sim} f, \quad i = 1, \ldots, n, \end{aligned} \tag{6}$$

where μ^{X_i} and Σ^{X_i} are the mean vector and covariance matrix that depend on X_i, and f is the distribution of the covariate over some space $\mathcal{X}$. Under the model in Equation (6), θ and δ admit a convenient analytical expression as expectations over the covariate distribution. Specifically, consider the following two assumptions:

- **Conditional Orthogonality of Potential Outcomes**:

$$\Sigma^x_{Y_1 Y_0} = \Sigma^x_{S_1 S_0} = 0 \quad \forall\, x \in \mathcal{X}.$$

- **Conditional Homoscedasticity**:

$$\Sigma^x_{Y_1Y_1} = \Sigma^x_{S_1S_1} = \Sigma^x_{Y_0Y_0} = \Sigma^x_{S_0S_0} = \frac{1}{2} \quad \forall\, x \in \mathcal{X}.$$

Then, one can show (see Appendix A) that

$$\delta = \mathbb{E}_{(X_i,X_j)\sim f\times f}(g(X_i,X_j))\,, \quad \theta = \mathbb{E}_{X_i\sim f}(g(X_i,X_i))\,, \tag{7}$$

where

$$g(x_i,x_j) = \Phi\left(\mu^{x_i}_{Y_1} - \mu^{x_j}_{Y_0}\right) - \Phi\left(\mu^{x_i}_{S_1} - \mu^{x_j}_{S_0}\right) \quad \forall\, x_i, x_j \in \mathcal{X}. \tag{8}$$

Without these assumptions, the formula for g still entails a difference of Gaussian CDFs, but with arguments involving the conditional variances and covariances, leading to a more cumbersome expression (see Appendix A).

Hence, θ and δ do not coincide whenever $\mathbb{E}_{X_j\sim f}\left(g(X_i,X_j)\right)$ and $g(X_i,X_i)$ differ in expectation. Importantly, this can happen even under the two more stringent assumptions stated above. To exemplify this, we consider two simple settings where the distribution of the potential outcomes is covariate-dependent.

2.4.1 Binary Covariate Setting

Let us start by considering the simplest possible setting where $f = \text{Bernoulli}\left(\frac{1}{2}\right)$. Although this choice may seem trivial, it can still have a meaningful interpretation in a clinical trial setting (e.g. X represents smoker vs non-smoker status). To illustrate the discrepancy between θ and δ (see Appendix B for the full derivation), suppose that the treatment effect is

homogeneous across all values of X:

$$\mu^x_{Y_1} - \mu^x_{Y_0} = \mu^x_{S_1} - \mu^x_{S_0} = \Delta, \quad x = 0, 1, \tag{9}$$

for some constant $\Delta > 0$. One can show (see Appendix B) that the discrepancy between θ and δ is driven by Δ and the difference between means of Y and S for control units across different X groups. Specifically, we have that

$$|\theta - \delta| = \frac{1}{4}|\Phi(\Delta + d_Y) - \Phi(\Delta + d_S) + \Phi(\Delta - d_Y) - \Phi(\Delta - d_S)|, \tag{10}$$

where $d_Y = \mu^0_{Y_0} - \mu^1_{Y_0}$ and $d_S = \mu^0_{S_0} - \mu^1_{S_0}$. Then, to understand how the discrepancy between θ and δ changes as a function of d_Y and d_S, it is useful to consider the plot of this quantity in Figure 1 for a fixed value of Δ. It appears that $|\theta - \delta|$ is very large whenever

$$|d_Y| < \Delta \text{ and } |d_S| > \Delta \quad \text{or} \quad |d_Y| > \Delta \text{ and } |d_S| < \Delta. \tag{11}$$

This implies that

$$\sup_{d_Y, d_S} \{|\theta - \delta|\} = \frac{1}{4}|1 - 2\Phi(\Delta)| \Rightarrow \sup_{d_Y, d_S, \Delta} \{|\theta - \delta|\} = \frac{1}{4}, \tag{12}$$

and, in particular, $\theta = 0$ while $|\delta|$ can be as large as $\frac{1}{4}$. This means that in this setting, even though S is a perfect surrogate for Y, δ is substantially different from 0, while θ correctly captures the surrogate validity.

2.4.2 Gaussian Covariate Setting

Let us now consider a setting where $f = \mathcal{N}(m, s^2)$ and the mean potential outcomes are linear functions of the covariate:

$$\mu^x = x\boldsymbol{\beta} \quad \forall\, x \in \mathbb{R}, \quad \text{where} \quad \boldsymbol{\beta} = (\beta_{Y_1}, \beta_{S_1}, \beta_{Y_0}, \beta_{S_0})^\top. \tag{13}$$

In general, δ and θ are non-trivial functions of the slope parameter $\boldsymbol{\beta}$ and the covariate distribution parameters m and s (see Appendix C for their closed-form expressions). Let us therefore impose the following assumption, which corresponds to perfect surrogacy:

$$\beta_{Y_1} - \beta_{Y_0} = \beta_{S_1} - \beta_{S_0} = \Delta, \tag{14}$$

which implies the treatment effect is the same for both Y and S across all values of the covariate. Under this assumption, $\theta = 0$ and therefore $|\theta - \delta| = |\delta|$, which is given by

$$|\theta - \delta| = \left| \Phi\left(\frac{m\Delta}{\sqrt{1 + s^2\left((\beta_{Y_0} + \Delta)^2 + \beta_{Y_0}^2\right)}} \right) - \Phi\left(\frac{m\Delta}{\sqrt{1 + s^2\left((\beta_{S_0} + \Delta)^2 + \beta_{S_0}^2\right)}} \right) \right|. \tag{15}$$

For $m, \Delta \neq 0$, the value of $|\theta - \delta|$ is entirely driven by the magnitude of $\boldsymbol{\beta}$. One can further show (see Appendix C) that

$$\sup_{\boldsymbol{\beta},\Delta}\{|\theta - \delta|\} = \left| \Phi\Big(\frac{m}{s}\Big) - \frac{1}{2} \right| \Rightarrow \sup_{\boldsymbol{\beta},\Delta,m,s}\{|\theta - \delta|\} = \frac{1}{2}, \tag{16}$$

where the outer supremum is approached for large $|m/s|$, that is, when the covariate is highly concentrated around a non-zero mean. This means that in this setting, even though S is a perfect surrogate for Y, δ can be substantially different from 0 and its value depends arbitrarily on the distribution of the covariate. In contrast, θ correctly captures the surrogate

validity by being equal to 0.

Both settings above illustrate that the choice of discrepancy parameter is crucial, as different choices can lead to very different conclusions about surrogate validity. The rank-based approach focuses on δ, which may not capture the causal relationship between Y and S even in simple settings. We therefore advocate focusing on θ instead.

3 Proposed Bayesian Imputation-Based Approach

The central idea of our approach is to estimate the parameter of interest and perform inference through its posterior distribution via Bayesian imputation. Specifically, we use the observed data to impute the unobserved potential outcomes from their posterior distribution. After imputing the missing potential outcomes, we can perform either between-unit or within-unit comparisons between treatment and control groups using any chosen discrepancy metric. Therefore, following the discussion in Section 2.3, we focus on the following quantities:

$$V_Y = \mathbb{P}(Y_{1i} > Y_{0i}), \quad V_S \ = \mathbb{P}(S_{1i} > S_{0i}), \quad \theta = V_Y - V_S. \tag{17}$$

As is typically done in Bayesian analysis, we perform inference on θ by sampling from its posterior distribution. At each iteration of our sampling algorithm, we impute the unobserved Y_{gi} and S_{gi} values (e.g., Y_{0i} and S_{0i} for treated units) from their posterior distribution. Using both the observed and imputed outcomes, we compute the probabilities in Equation (17) via Monte Carlo estimators, and obtain a posterior sample for θ by taking their difference. Then, using posterior draws of θ, we can perform inference and assess whether S is a valid surrogate. Specifically, we compute a one-sided credible interval for θ using a quantile of the

T posterior draws, after discarding the first b iterations as burn-in:

$$\widehat{\theta}_{1-\alpha} = \text{quantile}\Big(\widehat{\theta}^{(b+1:T)},\, 1-\alpha\Big)\,, \tag{18}$$

and perform the hypothesis test:

$$\begin{cases} H_0 : \theta \geq \eta, \\ H_1 : \theta < \eta. \end{cases} \tag{19}$$

where η is a pre-specified threshold for surrogate validation. We reject the null hypothesis if $\widehat{\theta}_{1-\alpha} < \eta$; otherwise, we fail to reject it. The full procedure is summarized in Algorithm 1.

The primary challenge of this methodology lies in sampling the unobserved potential outcomes. Clearly, for each unit i, the sampling strategy to draw the missing potential outcomes from their posterior distribution crucially depends on the assumed data-generating mechanism, denoted as F_i in Equation (1). Therefore, in the next section, we propose two flexible parametric models that can be used for this sampling step.

3.1 Parametric Models for the Data Generating Mechanism

The first model we consider excludes covariates, and therefore more closely aligns with the setting of Parast et al. (2024). Specifically, we define F_i from Equation (1) as follows:

$$\begin{aligned} P_i &\overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu, \Sigma)\,, \quad i = 1, \ldots, n, \\ \mu &\sim \mathcal{N}_4(\mu_0, \Sigma_0), \\ \Sigma &= \text{diag}(\sigma_{1:4})\,\Omega\,\text{diag}(\sigma_{1:4}), \\ \sigma_k &\overset{\text{i.i.d.}}{\sim} \text{Half-Normal}(0, s), \quad k = 1, 2, 3, 4, \\ \Omega &\sim \text{LKJ}(\tau), \end{aligned} \tag{20}$$

where we place a Gaussian prior on the mean vector μ and we adopt a separation strategy (Barnard et al., 2000) for the covariance matrix Σ, decomposing it into a correlation matrix Ω and a scale vector $\sigma_{1:4}$. This allows us to specify easily interpretable priors: an LKJ prior (Lewandowski et al., 2009) for the correlation and a half-normal prior for the scale.

However, one of the main advantages of our proposed Bayesian framework is its ability to naturally incorporate baseline covariates into the data-generating mechanism. To illustrate this feature, we consider a second model that introduces covariates through a linear regression structure, specifying F_i from Equation (1) as follows:

$$\begin{aligned}
&P_i \overset{\text{ind.}}{\sim} \mathcal{N}_4(BX_i, \Sigma)\,, \quad i = 1, \ldots, n, \\
&B = \left[\beta_1\ \beta_2\ \beta_3\ \beta_4\right]^\top, \\
&\beta_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}_d(\mu_\beta, \Sigma_\beta), \quad k = 1, 2, 3, 4, \\
&\Sigma = \operatorname{diag}(\sigma_{1:4})\,\Omega\,\operatorname{diag}(\sigma_{1:4}), \\
&\sigma_k \overset{\text{i.i.d.}}{\sim} \text{Half-Normal}(0, s), \quad k = 1, 2, 3, 4, \\
&\Omega \sim \text{LKJ}(\tau),
\end{aligned} \tag{21}$$

where we place a Gaussian prior on the regression coefficients in the matrix B and we adopt the same separation strategy for the covariance matrix Σ as in the first model.

3.2 Identifiability of θ and Interpretation of its Posterior

Both formulations in Equations (20) and (21) are quite simple and intuitive. However, they are both plagued by a common non-identifiability issue. Specifically, it is a well-known fact in the causal inference literature that the joint distribution of the potential outcomes is not identifiable from the observed data, since only one potential outcome is observed for each unit. This issue is typically referred to as the "fundamental problem of causal inference"

(Holland, 1986). Therefore, to understand how this issue manifests in our setting, it is useful to take a closer look at the correlation matrix Ω in the two models:

$$\Omega = \begin{bmatrix} 1 & \rho_{Y_1 S_1} & \rho_{Y_1 Y_0} & \rho_{Y_1 S_0} \\ \rho_{Y_1 S_1} & 1 & \rho_{S_1 Y_0} & \rho_{S_1 S_0} \\ \rho_{Y_1 Y_0} & \rho_{S_1 Y_0} & 1 & \rho_{Y_0 S_0} \\ \rho_{Y_1 S_0} & \rho_{S_1 S_0} & \rho_{Y_0 S_0} & 1 \end{bmatrix}. \tag{22}$$

It is easy to see that the green entries in the correlation matrix from Equation (22) are identifiable from the data, since they involve correlations between potential outcomes that are observed together in the data. In contrast, the remaining entries are not identifiable from the data, since they involve correlations between potential outcomes that are never observed together for any unit. Moreover, as we showed in Equation (7) for the Gaussian case, the red entries directly affect the computation of θ, while the orange entries do not.

In principle, one could try to address this non-identifiability issue by imposing further assumptions on the correlation matrix Ω. For example, one could assume that all the non-identifiable entries are equal to zero, which would imply that the potential outcomes are independent of each other. However, this assumption is quite strong and is likely unreasonable in many applications. But if we are not willing to make such strong assumptions, how can we still make sense of the posterior distribution of θ obtained from our Bayesian imputation approach?

Under standard regularity conditions — specifically, that the prior assigns positive mass to every KL-neighborhood of the truth and that the model is identifiable — the posterior concentrates around the true parameter value (Schwartz's theorem (Schwartz, 1965); see Appendix D for the formal statement and definition of regularity). Our model satisfies the first condition but fails identifiability, and therefore we cannot guarantee that the posterior

of θ concentrates around its true value. A characterization of what happens instead is given by Gustafson (2010):

Theorem 1. *Let the true data-generating process be defined as*

$$D_i \overset{i.i.d.}{\sim} f_{\phi^*}, \quad i = 1, \ldots, n, \tag{23}$$

where D_i is the observed data for unit i and f_{ϕ^*} is the true distribution of the data, parametrized by ϕ^*. Let the parameter ϕ be decomposed as follows:

$$\phi = (\phi_I, \phi_N), \tag{24}$$

where ϕ_I is the identifiable part of the parameter and ϕ_N is the non-identifiable part of the parameter. Let the Bayesian model be defined as

$$\begin{aligned} D_i \mid \phi &\overset{i.i.d.}{\sim} f_{\phi_I}, \quad i = 1, \ldots, n, \\ \phi &\sim \pi, \end{aligned} \tag{25}$$

where π is a prior distribution over the parameter ϕ. Then, if the identifiable component of the Bayesian model is regular according to Definition 1, the posterior distribution of ϕ concentrates around the set of values that are compatible with the observed data and the model assumptions as the sample size goes to infinity:

$$\pi\left(\phi_I \in U_I \mid D_{1:n}\right) \overset{P_{\phi^*}}{\to} 1, \tag{26}$$

for every neighborhood U_I of the identifiable component ϕ_I^*, where P_{ϕ^*} is the probability measure induced by the true distribution f_{ϕ^*}. Whereas, the posterior distribution of the

non-identifiable component ϕ_N will behave as follows:

$$\pi\left(\phi_N \in U_N \mid D_{1:n}\right) \overset{P_{\phi^*}}{\to} \pi\left(\phi_N \in U_N \mid \phi_I = \phi_I^*\right), \tag{27}$$

for every neighborhood U_N of the non-identifiable component ϕ_N^*, where $\pi\left(\phi_N \in U_N \mid \phi_I = \phi_I^*\right)$ is the conditional prior distribution of the non-identifiable component given the identifiable component evaluated at its true value.

What this last result tells us is that, even though the posterior distribution of θ is not fully identified from the data, there is still a process of Bayesian learning that takes place as the sample size increases, and the posterior distribution of θ will concentrate around a set of values that are compatible with the observed data and the model assumptions. Therefore, it is still possible to meaningfully perform inference on θ for the purpose of surrogate validation. However, it is important to keep in mind that due to the lack of concentration for the non-identifiable component of the parameter, there is a non-vanishing uncertainty in the posterior distribution of θ that persists even in the asymptotic limit. This feature of the proposed Bayesian imputation approach may appear to constitute an inherent limitation. Nevertheless, we argue that the methodology retains merit for at least two reasons.

First, the identifiability issue is not unique to our proposed Bayesian imputation approach, but rather it is a fundamental issue that arises in any approach that relies on the observed data to make inference about the joint distribution of the potential outcomes. Therefore, any method that attempts to assess surrogate validity using the observed data will have to contend with this issue in some form. Then, the non-vanishing uncertainty in the posterior distribution of θ can be seen as a reflection of the inherent uncertainty in the problem of surrogate validation. However, since the amount of uncertainty injected by the prior does not disappear even asymptotically, it becomes of crucial importance to choose reasonable

priors that reflect our beliefs about the data-generating process. In this regard, given the prior specifications chosen in Equations (20) and (21) and, more specifically, the choice of the prior for the correlation matrix Ω, unless we have strong prior beliefs about the correlation structure of the potential outcomes, it is reasonable to choose a weakly informative prior, such as an LKJ prior with $\tau = 1$, which corresponds to a uniform distribution over the space of correlation matrices.

Second, even though our proposed Bayesian imputation approach does not showcase the desirable property of consistency, it still provides useful information about surrogate validity and, most importantly, it may still be preferable to alternative approaches that may not suffer by this non-identifiability issue but are unable to correctly capture the causal relationship between the surrogate and the primary outcome. For example, as we showed in Section 2.3, the discrepancy parameter δ used by Parast et al. (2024) may fail to capture the causal relationship between Y and S even in very simple settings, while our approach targets a parameter θ that is able to correctly reflect surrogate validity. Hence, this suggests that using our approach can lead to more accurate assessments of surrogate validity than the rank-based approach of Parast et al. (2024) and especially in small-sample settings where the lack of asymptotic consistency may not be as problematic.

3.3 Selection Procedure for the Validation Threshold

We now turn to the challenge of selecting the prespecified threshold used to assess surrogate validity, denoted ε by Parast et al. (2024) and η in Equation (19). As noted by Parast et al. (2024), the distribution of $\widehat{\delta}$ is not fully specified under the null in Equation (3), so choosing the threshold ε is not a trivial task. They suggest choosing it to ensure a desired power to detect a treatment effect on S. Here, we outline an alternative procedure for selecting η in Equation (19), similar in spirit but better suited to our Bayesian framework, which relies on

a Bayes factor, the standard tool for Bayesian hypothesis testing.

The intuition is that we aim to set η such that when V_S is close to V_Y i.e., within η, S is a valid replacement of Y. To do this, we aim to identify v_s, which is defined as the value of V_S that ensures a power of $(1-\beta)100\%$ for the following hypothesis test:

$$\begin{cases} H_0 : V_S = \frac{1}{2}, \\ H_1 : V_S > \frac{1}{2} \end{cases} \tag{28}$$

implemented using a Bayes factor. Let v_Y denote the hypothesized treatment effect on Y. One reasonable way to choose v_Y is by using the posterior mean of V_Y obtained from our Bayesian imputation framework. Then $v_Y - v_S$ reflects how far we are willing to let v_S be from v_Y, while still ensuring a $(1-\beta)100\%$ power. To this end, the value of η is defined as:

$$\eta = \max\{v_Y - v_S, 0\}. \tag{29}$$

We now describe the details behind obtaining the value v_s. To compute a Bayes factor that can be used to conduct the hypothesis test described in Equation (28), we first need to specify a Bayesian model taking V_S as its unknown parameter:

$$n\,\widehat{V}_S \mid V_S \sim \text{Binomial}(n, V_S), \quad \text{and} \quad V_S \quad \sim p(V_S), \tag{30}$$

where the prior for V_S is a Beta distribution renormalized on $\left(\frac{1}{2}, 1\right)$:

$$p(V_S) = \frac{\text{Beta}(V_S \mid a, b)}{\int_{\frac{1}{2}}^{1} \text{Beta}(v \mid a, b)\, dv} \;\forall\, V_S \in \left(\tfrac{1}{2}, 1\right), \tag{31}$$

with $a, b > 0$ suitably chosen prior parameters, and $\widehat{V}_S = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(S_{1i} > S_{0i})$. Note that $\widehat{V}_S$ cannot be computed directly from the observed data, since S_{1i} and S_{0i} are never jointly

observed for any unit i. However, as we shall see below, this is not an issue, because the procedure to select v_s only requires the sampling distribution of $\widehat{V}_S$ under the null hypothesis and alternative hypothesis, rather than an observed value for it.

The Bayes factor arising from this model is (see Appendix E for the derivation):

$$BF_n = \frac{B\Big(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\Big)\,\Big(1 - F_{\mathrm{Beta}\left(a+n\,\widehat{V}_S,\, b+n-n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\Big)}{B(a,b)\,\Big(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\Big)} \cdot 2^n, \tag{32}$$

where $B(a,b)$ is the Beta function and $F_{\mathrm{Beta}(a,b)}$ is the CDF of the $\mathrm{Beta}(a,b)$ distribution. As usual, we would like to ensure a type I error probability of α:

$$\mathbb{P}(BF_n > BF_{n,\alpha} \mid H_0) = \alpha. \tag{33}$$

Since $n\,\widehat{V}_S$ takes values in $\{0,\ldots,n\}$, BF_n has at most $n+1$ distinct values, so $BF_{n,\alpha}$ can be obtained as the $(1-\alpha)$-quantile of BF_n (see Appendix E for the closed-form distribution of BF_n under H_0). Given $BF_{n,\alpha}$, we can define v_S as the value such that

$$\mathbb{P}\left(BF_n > BF_{n,\alpha} \mid V_S = v_S > \frac{1}{2}\right) = 1 - \beta. \tag{34}$$

It useful to observe that $\forall\, v_S \in \left(\frac{1}{2}, 1\right)$,

$$v_S \mapsto \mathbb{P}\big(BF_n > BF_{n,\alpha} \mid V_S = v_S\big) \tag{35}$$

is a monotone function. Therefore, there exists a unique v_S satisfying Equation (34). The probability can be evaluated using the same procedure described in Appendix E for the distribution of BF_n given $V_S = v_S$. Since the function is continuous and monotone, v_S can be obtained using standard numerical root-finding algorithms, such as `uniroot` in `R`.

4 Simulation Study

We evaluated the performance of our proposed Bayesian imputation-based method against the frequentist rank-based approach of Parast et al. (2024) in terms of test coverage and power via a simulation study in six settings.

Specifically, in Settings 1 to 4, we replicated the data-generating mechanisms used by Parast et al. (2024) to facilitate direct comparison. In Settings 1 to 3 the potential outcomes for the primary outcome were generated as Gaussian variables, but in setting 1 the surrogate was designed to be useless, since its potential outcomes were generated independently of the primary outcome, while in settings 2 and 3 the potential outcomes for the surrogate were obtained by applying a linear transformation of the primary outcome, with the addition of Gaussian noise. Moreover, in Setting 2, the surrogate was designed to be perfect, since the noise was negligible, while in Setting 3, the surrogate was designed to be imperfect, since the noise was more substantial. In contrast, in Setting 4 the outcome was generated from a non-Gaussian distribution and the surrogate was generated by applying a non-linear transformation with the addition of non-Gaussian noise. In all of these settings, we implemented the model in Equation (20) without covariates. Given the true data generating mechanism, one would expect our proposed method to work well in Settings 1-3, but possibly not work well in Setting 4.

In Settings 5 and 6, we purposefully aimed to break the frequentist rank-based method and highlight the advantage of our proposed approach based on the model in Equation (21) with covariates. Setting 5 features a binary baseline covariate, with generating parameters chosen to maximize the discrepancy between δ and θ, which are approximately 0.25 and 0, respectively. Conditionally on the covariate, the potential outcomes are not identically distributed across units, violating a key assumption of the Mann-Whitney test. In this setting the surrogate is a replica of the primary outcome with the addition of Gaussian noise,

with treatment effect homogeneity across the two subpopulations and uncorrelated potential outcomes; therefore, any valid surrogate validation method should correctly identify S as valid. Setting 6 features a Gaussian baseline covariate with potential outcomes following a linear Gaussian model. The conditional treatment effect on Y and S is identical (implying $\theta = 0$), but the marginal rank-based discrepancy δ is nonzero due to the covariate structure, further illustrating the limitation of rank-based methods in covariate-dependent settings. Full details of all data-generating processes, prior parameters, and MCMC settings are given in Appendix F.

We ran 500 simulations for each setting, with a sample size of $n = 50$ and a significance level of $\alpha = 0.05$. We quantify performance in terms of coverage of the confidence/credible intervals of the true parameter (δ for the frequentist method and θ for the Bayesian method) and power, by which we mean the proportion of simulation iterations where the method deemed the surrogate to be valid. In Setting 1, the surrogate is not valid, and thus, this is the Type 1 error rate.

Results are shown in Table 1. In Settings 1 and 2, the rank-based method of Parast et al. (2024) and the Bayesian imputation-based method perform similarly. In in Setting 3,improved performance is observed for the Bayesian method with higher coverage and higher power. As expected, in Setting 4, where the data-generating process is non-Gaussian and non-linear, the nonparametric rank-based method outperforms the Bayesian method in terms of power. Settings 5 and 6 highlight the true strength of our proposed Bayesian imputation-based method, where the frequentist method very rarely identifies the surrogate as valid, while our method correctly recognizes the surrogate as valid in all iterations. This is not surprising, since the data-generating process in Settings 5 and 6 were purposely designed to illustrate the limitations of the the rank-based method. The proposed Bayesian method is able to adjust for the binary covariate in these settings, and correctly capture surrogacy.

The results demonstrate the power gains of our proposed methodology, though we also intentionally explore its limitations under model misspecification.

5 Application

We illustrate our proposed method using data from the Diabetes Control and Complications Trial (DCCT). This trial was a multicenter, randomized, clinical study designed to determine the effect of an intensive treatment protocol directed at maintaining blood glucose concentrations on progression of early vascular complications in patients with Type 1 diabetes, compared to a control group that received usual care (DCCT, 1987; Lachin and Braffett, 2021; DCCT, 1993). Specifically, the intensive therapy was designed to achieve normal blood glucose levels with three or more insulin injections or an insulin pump. Usual care consisted of 1 or 2 insulin injections per day. Our primary outcome of interest was change in hemoglobin A1c (HbA1c) from baseline to 4.5 years years post-randomization, while the surrogate was an earlier measurement of change in HbA1c from baseline to 1.5 years post-randomization.

Our analysis is focused on a small subgroup of high-risk participants, specifically those in the secondary-intervention cohort (with baseline mild-to-moderate retinopathy) with a body mass index over 25.0 with a history of smoking. This resulted in an analytic sample of 47 participants. Figure 2 describes the data, showing the distribution of the primary outcome and the surrogate (top panel), as well as the relationship between the surrogate and the primary outcome (bottom panel) in the treated and control groups within this high-risk subgroup.

We first applied our proposed method without covariates as described above, with a specified power of 0.70 used to determine the corresponding threshold. The resulting posterior

distribution of $\widehat{\theta}$ is shown in the left panel of Figure 4, with an average estimate of 0.027, and a credible interval upper bound of 0.149. The calculated threshold was 0.164 and thus, we conclude that change in HbA1c from baseline to 1.5 years is a surrogate for change in HbA1c from baseline to 4.5 years post-randomization. Next, we applied our proposed procedure incorporating the covariates of age (numeric) and sex (male or female). The resulting posterior distribution of $\widehat{\theta}$ incorporating these covariates is shown in the right panel of Figure 4, with an average estimate of 0.010, and a credible interval upper bound of 0.149. The calculated threshold was 0.158 and thus, we similarly conclude that this is a valid surrogate. For comparison, the method of Parast et al. (2024) without covariates (because covariates are not possible to incorporate) resulted in $\widehat{\delta} = 0.044$, a confidence interval upper bound of 0.120, and with a threshold of 0.132, concluding that this is a surrogate marker. While the conclusions are similar, it is certainly worth noting that the estimated difference quantities ($\widehat{\theta}$ vs $\widehat{\delta}$) are smaller using the proposed definition and estimation approaches. Of course, the true values and true surrogacy here are unknown. Nonetheless, this application highlights the potential utility of our proposed approach and, in particular, demonstrates improved characterization of surrogacy and increased efficiency compared to the existing approach.

6 Discussion

We proposed a Bayesian imputation framework for surrogate evaluation in randomized clinical trials. Our method overcomes limitations of the frequentist rank-based approach, including lack of causal interpretation, inability to use baseline covariates, and low power, particularly in in covariate-dependent settings. By imputing unobserved potential outcomes, it enables flexible treatment-control comparisons and inference on surrogate-primary associations. Incorporating baseline covariates improves applicability to complex scenarios, as

shown in simulations where our method achieved perfect power versus 0.075 for the existing approach. Our proposed methods are implemented in the R package `BSET` available at `https://github.com/PietroCarlotti/BSET`, which also contains all reproducible code.

While our approach overcomes some limitations of the rank-based approach, there is room for further improvement, which we are actively pursuing. A key limitation is the reliance on parametric assumptions about the data-generating mechanism, which are vulnerable to model mis-specification. One possible direction, which we plan for future work, is to relax these assumptions by extending the framework to a Bayesian nonparametric setting (Ghosal and Van der Vaart, 2017), allowing a data-driven characterization of both the surrogate-outcome relationship and the covariate-outcome association. However, this flexibility comes at a cost of increased computational complexity and larger required sample sizes (Miller and Harrison, 2018). Finally, sensitivity analyses examining robustness of our conclusions to unidentified covariance parameters would also be useful, and we aim to incorporate such results into our framework and future software.

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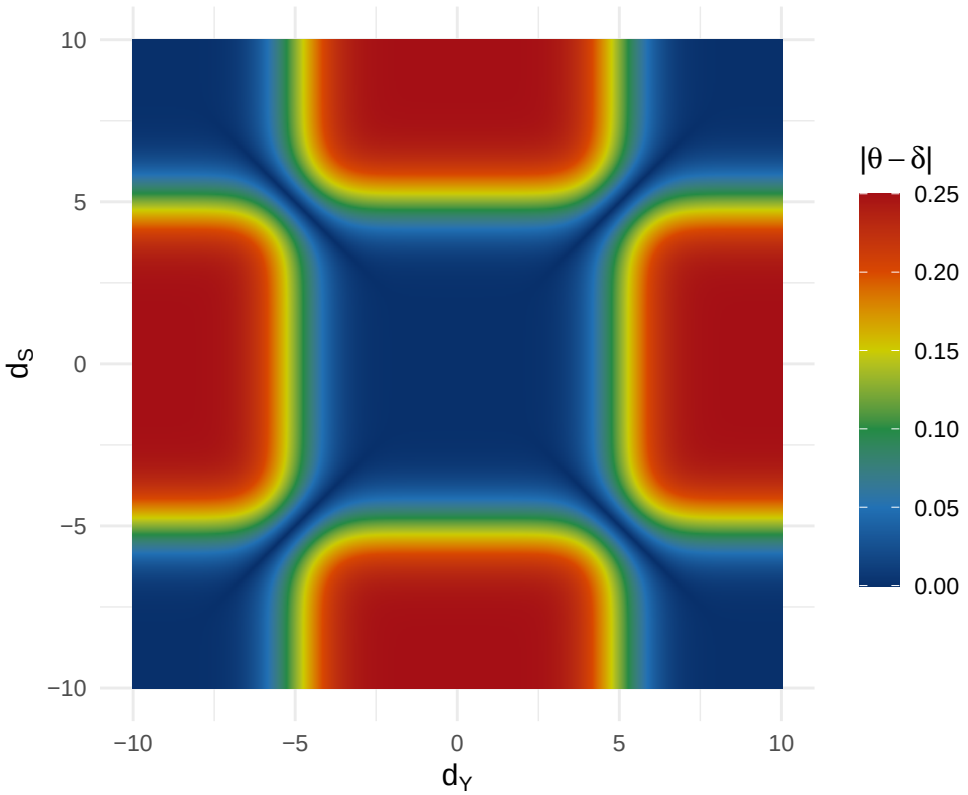


Figure 1: Heatmap of $|\theta - \delta|$ as a function of d_S and d_Y with $\Delta = 5$.

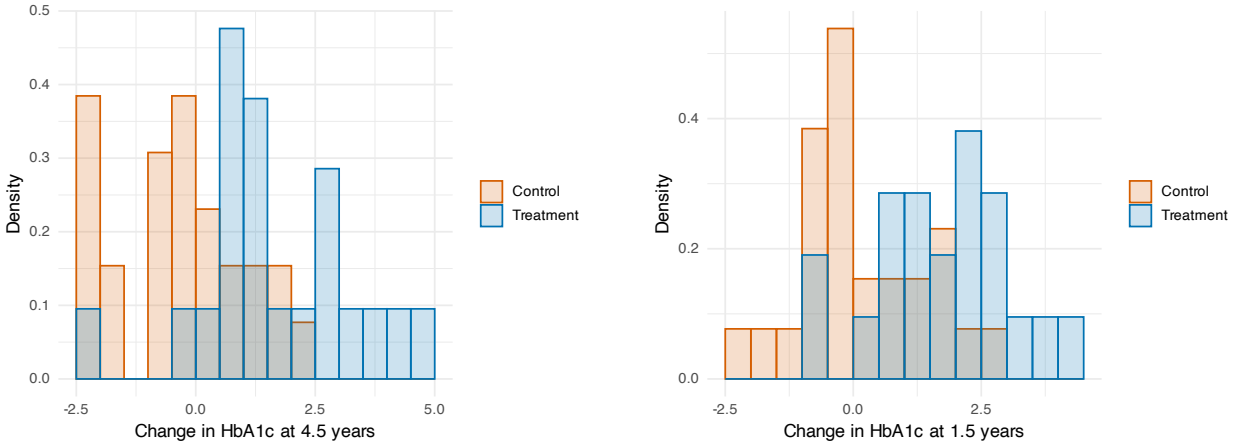
Table 1: Simulation results comparing the frequentist rank-based method and the proposed Bayesian imputation-based method in terms of coverage and power in Settings 1–5, summarized over 500 replications with $n = 50$ per replication. Bolding indicates the better performing method.

Setting	**Valid Surrogate**	**Coverage**		**Power**	
		Frequentist	Bayesian	Frequentist	Bayesian
1	No	0.98	**1.00**	0.00	0.00
2	Yes	0.98	**1.00**	1.00	1.00
3	Yes	0.90	**0.98**	0.54	**0.64**
4 (Misspecified)	Yes	0.96	**1.00**	**0.56**	0.06
5 (Binary Covariate)	Yes	0.925	**0.975**	0.075	**1.00**
6 (Gaussian Covariate)	Yes				

Algorithm 1 Bayesian Imputation-Based Approach.

Require: Observed data $\{Y_i, S_i, Z_i, X_i\}_{i=1}^n$, prior parameters, number of posterior iterations T, burn-in b, surrogate threshold η, significance level α

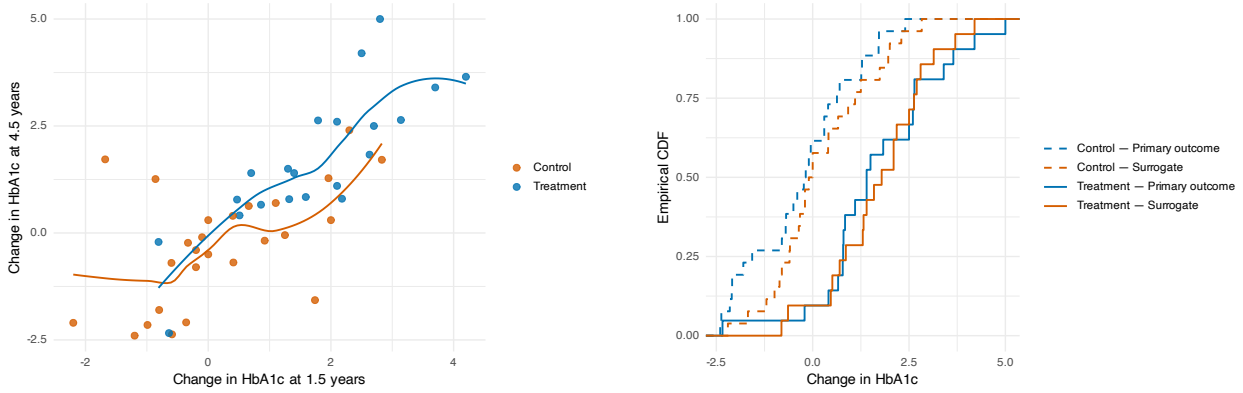
```
for t = 1 to T do
    for i = 1 to n do
        if Z_i = 1 then
            Sample P_{0i}^{(t)} ~ p(P_{0i}^{(t)} | P_{1i}, X_i, Z_i = 1)
        else
            Sample P_{1i}^{(t)} ~ p(P_{1i}^{(t)} | P_{0i}, X_i, Z_i = 0)
        end if
    end for
    V̂_Y^{(t)} ← (1/n) Σ_{i=1}^n 𝟙(Y_{1i}^{(t)} > Y_{0i}^{(t)})
    V̂_S^{(t)} ← (1/n) Σ_{i=1}^n 𝟙(S_{1i}^{(t)} > S_{0i}^{(t)})
    θ̂^{(t)} ← V̂_Y^{(t)} − V̂_S^{(t)}
end for
θ̂_{1−α} ← quantile(θ̂^{(b+1:T)}, 1 − α)
if θ̂_{1−α} < η then
    Reject H_0
else
    Fail to reject H_0
end if
```



(a) Primary outcome

(b) Surrogate

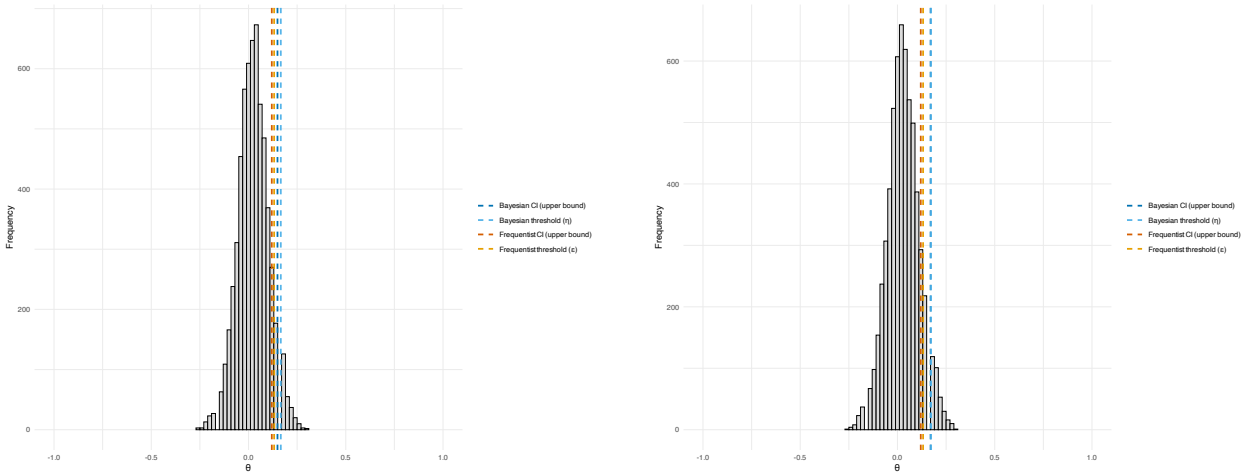
Figure 2: Diabetes Control and Complications Trial (DCCT) data illustration of the distribution of the primary outcome of interest, change in hemoglobin A1c (HbA1c) from baseline to 4.5 years post-randomization (left panel), and the surrogate, change in HbA1c from baseline to 1.5 years post-randomization (right panel), in the treated and control groups within the high-risk subgroup.



(a) Surrogate vs. primary outcome

(b) Empirical CDFs

Figure 3: Left panel: scatter plot of the surrogate (change in HbA1c at 1.5 years) against the primary outcome (change in HbA1c at 4.5 years) by treatment arm, with LOESS curves. Right panel: empirical cumulative distribution functions of the primary outcome and the surrogate by treatment arm. The horizontal gap between the solid and dashed curves of the same color reflects the treatment effect on each outcome, directly corresponding to the rank-based quantities U_Y and U_S.



(a) Without covariates

(b) With covariates (sex and BMI)

Figure 4: Posterior distributions of $\widehat{\theta}$ when applying the method with no covariates (left panel) and with age and sex (right panel) to the Diabetes Control and Complications Trial (DCCT) data where the primary outcome of interest was change in hemoglobin A1c (HbA1c) from baseline to 4.5 years post-randomization and the surrogate was change in HbA1c from baseline to 1.5 years post-randomization.